

The $\mathbb{Z}'_{31}$ – Ω_{32} Cosmological Engine

A Geometric Framework for Phase, Matter, Chirality, Color, and Time

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Abstract

I propose a new theoretical model of the universe. The physical universe is modeled as a discrete phase engine built on the cyclic group $\mathbb{Z}'_{31}$, where each $+8 \pmod{31}$ increment represents the fundamental tick of a topological $U(1)$ phase. Here $\mathbb{Z}'_{31}$ denotes the 31-state discrete $U(1)$ phase lattice, distinguished from the ordinary integer ring by its physical interpretation as a topological phase engine. This engine is embedded in a 31-state cubic propagation geometry consisting of 6 faces, 12 edges, 8 vertices, 4 body diagonals, and 1 center. The cube's edges support a natural 3-color partition ($SU(3)$ -like), while the diagonal–vertex overlap supports $U(1)_{EM}$. The universe evolves through a washboard potential dominated by a $21/31$ Dark Energy tilt acting across an $8/31$ Dark Matter scaffolding, forcing the phase to ratchet forward and generating the Arrow of Time, entropy, chirality, and matter–antimatter asymmetry as geometric consequences of the coprime structure. All particles, forces, and temporal behavior emerge from this geometric topological computation.

1 $\mathbb{Z}'_{31}$ as a Discrete $U(1)$ Phase Lattice

The universe's fundamental degree of freedom is a 31-state phase ring: $\mathbb{Z}'_{31}$. Its generator, $+8 \pmod{31}$, is the irreducible clock tick of reality. Because 8 and 31 are coprime, repeated application of $+8$ cycles through all 31 states without repetition, preventing phase locking and ensuring perpetual evolution.

- CW rotation $(+8) \rightarrow$ forward computation (matter)
- CCW rotation $(-8) \rightarrow$ reverse computation (antimatter)

These two directions encode helicity, chirality, and electromagnetic charge. Time is the sequential evaluation of this phase.

2 The 31-State Spatial Decomposition

The $\mathbb{Z}'_{31}$ engine is embedded in a cubic propagation graph: $31 = 6 + 12 + 8 + 4 + 1$.

- 6 faces $\rightarrow$ planar propagation
- 12 edges $\rightarrow$ linear propagation

- 8 vertices $\rightarrow$ point-source diagonal propagation
- 4 body diagonals $\rightarrow$ long-range waveguides
- 1 center $\rightarrow$ origin engine

3 SU(3)-Like Color on the Edges

The cube graph is 3-regular, allowing a perfect 3-edge-coloring. The 12 edges partition into 3 groups of 4: C_1, C_2, C_3 . Each vertex touches one edge of each color, serving as a color junction. This edge-coloring is unique up to permutation of the three colors. It is the geometric analog of SU(3) color. In accordance with the discrete analog of the Wu-Yang dictionary, the parallel transport of the phase along these edges functions as a topological gauge field, satisfying local gauge invariance organically without the need for continuous differential manifolds.

4 Electromagnetism on the Diagonal–Vertex Overlap

The 4 body diagonals and 8 vertices form the EM substructure.

- Diagonals $\rightarrow U(1)_{EM}$ waveguides
- Vertices $\rightarrow$ charge sites

Each vertex lies on exactly one body diagonal. The $\mathbb{Z}'_{31}$ phase is the EM phase: CW evolution $\rightarrow$ positive charge; CCW evolution $\rightarrow$ negative charge. The dual role of vertices as both color junctions and charge sites is intentional — a quark carries both color and electromagnetic charge simultaneously. Furthermore, because exactly three color-edges and one diagonal intersect at each vertex, the vertex transitions inherently satisfy the Yang-Baxter Equation ($R_{12}R_{13}R_{23} = R_{23}R_{13}R_{12}$). This ensures that the topological computation of color and charge is exactly solvable, protecting the invariant identity of the fermion under the sequence of edge-scattering.

5 Matter and Antimatter as CW vs CCW Phase

Because $\mathbb{Z}'_{31}$ is a $U(1)$ phase lattice: CW = ψ = matter; CCW = ψ^* = antimatter. The global structure favors CW, driven by the washboard tilt of §6, suppressing CCW except in local retrograde events, reproducing matter dominance and CP asymmetry as geometric consequences of the coprime structure.

6 The Washboard Potential and the 21/31 Tilt

The universe is dominated by the 21/31 Dark Energy fraction, an unbound phase rolling downhill across the discrete 8-fold lattice (the 8/31 Dark Matter scaffolding). These fractions arise from the Ω_{32} null state boundary structure: 8 states are captured into the bound scaffolding and 21 remain free to roll. This creates a macroscopic tilt: $+8 \pmod{31}$ is downhill; $-8 \pmod{31}$ is globally forbidden. This tilt is the Arrow of Time.

7 Time as Sequential Modulo-31 Computation

Time is the macroscopic measurement of the universe sequentially calculating its own modulo-31 topology. Each +8 step is a topological clock tick. The universe does not move through time — it computes time.

8 Entropy as Modulo Friction

Because the $U(1)$ phase generator (8) and the cubic topology (31) are coprime, the engine cannot achieve static resonance and must perpetually cycle. Entropy is the mechanical accumulation of topological friction generated by this unavoidable phase-step remainder as the continuous current is forced through the discrete 8-fold lattice. Entropy is therefore deterministic, not probabilistic — the macroscopic observation of constant geometric exhaustion.

9 Chirality and the Right-Handed Universe

The derivative structure of $\mathbb{Z}'_{31}$ preserves CW and suppresses CCW, eliminating CCW entirely after second derivation. This yields a right-handed chiral universe, matching the Standard Model. This geometric suppression provides a deterministic, mechanical origin for the weak parity violation observed by Lee and Yang, moving it from an empirical observation to a structural necessity.

10 C, P, T, CP, CPT in the $\mathbb{Z}'_{31}$ Engine

- **C (Charge Conjugation):** $CW \leftrightarrow CCW$ (Matter $\leftrightarrow$ Antimatter)
- **P (Parity):** Spatial inversion of the cube geometry
- **T (Time Reversal):** Globally forbidden by the 21/31 tilt
- **CP:** Broken by the washboard tilt (natural CP violation)
- **CPT:** Locally exact because the $\mathbb{Z}'_{31}$ engine is algebraically symmetric under combined phase conjugation, spatial inversion, and ratchet reversal — but globally broken by cosmological tilt

11 Closing: Expansion and Moore Neighborhood Re-seeding

Each $\mathbb{Z}'_{31}$ engine, upon completing its internal 31-state cycle, projects outward into the surrounding 3D Moore Neighborhood of 26 adjacent cells. The original cell seeds its neighbors with identical 2/31 fermionic boundary states, ensuring that the geometric structure remains constant across expansion. This reseeded process propagates the phase engine outward through the Ω_{32} ground state, the 32-state fermionic null-state reservoir surrounding each engine. Cosmic expansion is therefore the iterative tiling of the $\mathbb{Z}'_{31}$ engine, not the stretching of spacetime — a recursive replication of the same chirality, color structure, and EM coupling across the infinite lattice.

Concluding Remarks on Effective Geometry

The cubic propagation graph and $\mathbb{Z}'_{31}$ engine presented here are not necessarily an ontological claim about the literal, microscopic shape of space. Rather, they serve as a discrete computational substrate. This framework posits an isomorphism between the observable universe and the $\mathbb{Z}'_{31}$ - Ω_{32} engine. Whether the universe is fundamentally a discrete cubic lattice, or simply behaves in a manner mathematically indistinguishable from one, the predictive power of the coprime generator remains the same. By treating the null state as a topological structure rather than a continuous manifold, we extract a parameter-free resolution to established physical mechanics without the need for hidden continuous dimensions.

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